Torrent v0.66.0 — Benchmark Report & Performance Analysis

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Overview

This report presents benchmark results comparing **torrent v0.66.0** against **rqbit v8.1.1** and **qbittorrent-nox v5.1.4** (backed by libtorrent-rasterbar). All three clients downloaded the Arch Linux ISO (~1.5 GiB) from a public magnet link over 5 trials on the same machine and network.

The report also catalogues the performance bottlenecks discovered through code-level analysis of the torrent codebase, prioritised by their impact on **download throughput (MB/s)** and **memory usage**.

Benchmark Configuration

Parameter	Value
Torrent	`archlinux-2026.03.01-x86_64.iso` (1.5 GiB)
Trials	5 per client
Metrics	Wall time, CPU time (user+sys), avg download speed, peak RSS
Measurement	`/usr/bin/time -v` wrapper
Machine	CachyOS (Arch-based), Linux 6.19.0-2-cachyos

Benchmark Results

Summary Table (mean ± stddev, 5 trials)

Client	Time (s)	CPU (s)	Speed (MB/s)	RSS (MiB)
torrent 0.66.0	100.3 ±19.3	37.1 ±4.7	15.0 ±3.5	170.2 ±42.9
rqbit 8.1.1	34.4 ±24.1	10.4 ±0.6	56.7 ±27.0	37.5 ±3.9
qbittorrent 5.1.4	51.5 ±10.7	18.1 ±2.5	29.1 ±5.3	1525.5 ±5.5

Per-Trial Data

Client	Trial	Time (s)	CPU (s)	Speed (MB/s)	RSS (MiB)
torrent	1	117.7	39.7	12.3	154.8
torrent	2	100.1	39.4	14.5	184.1
torrent	3	115.6	35.9	12.6	135.4
torrent	4	98.6	41.3	14.7	238.6
torrent	5	69.4	29.5	20.9	138.2
rqbit	1	18.6	10.7	78.3	39.9
rqbit	2	19.7	9.5	73.6	32.5
rqbit	3	19.1	10.1	75.8	34.1
rqbit	4	40.1	10.8	36.2	39.9
rqbit	5	74.4	10.8	19.5	41.1
qbittorrent	1	44.0	18.9	33.0	1534.6
qbittorrent	2	49.8	17.3	29.1	1524.9
qbittorrent	3	41.8	14.7	34.7	1525.7
qbittorrent	4	68.9	21.6	21.1	1521.1
qbittorrent	5	52.8	18.1	27.5	1521.1

Relative Performance (torrent vs rqbit)

Metric	torrent	rqbit	Ratio
Download speed	15.0 MB/s	56.7 MB/s	**3.8x slower**
Peak RSS	170.2 MiB	37.5 MiB	**4.5x more memory**
CPU time	37.1s	10.4s	3.6x more CPU

Relative Performance (torrent vs qbittorrent)

Metric	torrent	qbittorrent	Ratio
Download speed	15.0 MB/s	29.1 MB/s	**1.9x slower**
Peak RSS	170.2 MiB	1525.5 MiB	**9x less memory** (win)
CPU time	37.1s	18.1s	2x more CPU

Key Observations

- torrent's `final_speed=0.0 MB/s` on all 5 trials — speed drops to near zero in the last 5–10% of the download, adding 40–60 seconds per run
- torrent hits the 200-peer cap on all trials but cannot sustain throughput
- rqbit trials 4 and 5 show swarm variability (36 and 19 MB/s vs typical 75+ MB/s), indicating shared network conditions affect all clients equally
- qbittorrent uses 1.5 GiB RSS — libtorrent-rasterbar's memory-mapped I/O model is inherently memory-heavy
- torrent's only competitive advantage is memory efficiency vs qbittorrent

Performance Bottlenecks — Prioritised by Impact

The following bottlenecks were identified through code-level analysis comparing torrent's architecture against rqbit. They are ranked by their expected impact on **download throughput** and **memory usage** (CPU overhead is deprioritised per project goals).

Priority 1 — End-Game Pipeline Starvation (Throughput)

Location: `torrent.rs:5364`

Problem: `END_GAME_DEPTH = 4` limits each peer to 4 in-flight requests during end-game mode. Normal mode uses 128. With 50 peers at 100ms RTT:

Mode	Calculation	Throughput Ceiling
Normal	50 × 128 × 16KB / 0.1s	~1 GB/s
End-game	50 × 4 × 16KB / 0.1s	~32 MB/s

The download spends 5–10% of its time in end-game, but that translates to **40–60 seconds of wall time** (the tail from ~15 MB/s down to 0 MB/s). This is the single largest bottleneck.

rqbit's approach: Flat 128-permit semaphore, no depth reduction in end-game.

Fix: Raise `END_GAME_DEPTH` to 128 (match normal mode).

Priority 2 — Reactive Re-Requesting Cascade (Throughput)

Location: `torrent.rs:3601-3627`

Problem: When a block is received in end-game mode, duplicate requests are cancelled across peers. For each cancelled peer, the code immediately invokes a **full picker cycle** (`request_end_game_block()`). Then it invokes *another* picker cycle for the receiving peer. With 50 peers in end-game:

- Each block → 2+ full picker cycles
- At 6,000 blocks/sec → **18,000 picker invocations/sec**
- Optimal rate: ~50 (one per peer per second during refill)

Each picker cycle iterates all pieces, clones bitfields, and scans in-flight state. This creates a CPU bottleneck that caps throughput even when network bandwidth is available.

rqbit's approach: Simple block tracking — remove from pending, no reactive cascade.

Fix: Batch freed peers into the next tick's refill cycle instead of triggering immediate picker cycles.

Priority 3 — Unbounded Store Buffer Memory (Memory)

Location: `disk.rs:47-52`

Problem: Every in-flight piece maintains a `BTreeMap<u32, Bytes>` holding all received blocks in RAM until piece verification completes:

```
type StoreBuffer = Mutex<HashMap<(Id20, u32), BTreeMap<u32, Bytes>>>
```

With 100+ pieces in-flight × 32 blocks/piece × 16 KB/block = ~**50 MiB** of buffered data. This is the primary contributor to the 170 MiB vs 37.5 MiB RSS gap.

rqbit's approach: Writes blocks directly to disk via `pwritev()`, verifies from on-disk data. No in-memory buffer.

Fix: Write blocks to disk immediately, verify from disk. Or cap in-flight buffer to N pieces and apply backpressure.

Priority 4 — O(n) Linear Scans in pending_requests (Throughput)

Location: `torrent.rs:3352, 3520, 3583, 3619` (4 call sites)

Problem: `pending_requests` is a `Vec<(u32, u32, u32)>` searched with `.iter().position()` on every block received or cancelled. With 128 entries per peer and 6,000 blocks/sec:

- 128-entry linear scan × 6,000/sec = **768,000 comparisons/sec**
- In end-game, cancel propagation multiplies this across peers

rqbit's approach: Indexed by `(piece, begin)` hash for O(1) lookup.

Fix: Replace `Vec` with `FxHashMap<(u32, u32), u32>` keyed on `(piece_index, block_offset)`.

Priority 5 — peer_count() Sort-and-Dedup (Throughput)

Location: `piece_selector.rs:73-81`

Problem: Counting unique peers assigned to a piece collects all values into a Vec, sorts, and dedups — O(n log n) per call. Called for every in-flight piece during each peer's pick cycle.

```rust
let mut peers: Vec<SocketAddr> = self.assigned_blocks.values().copied().collect();
peers.sort_unstable();
peers.dedup();
peers.len()
```

With 100+ pieces and 50+ peers, this runs thousands of times per second.

Fix: Maintain a `HashSet<SocketAddr>` or a simple counter that increments/decrements as blocks are assigned/completed.

Priority 6 — Redundant Bitfield Clones in Hot Path (Memory + Throughput)

Location: `torrent.rs:5375`

Problem: Peer bitfield is cloned on every iteration of the picker loop:

```rust
let peer_bitfield = match self.peers.get(&peer_addr) {
    Some(p) => p.bitfield.clone(),
    None => return,
};
```

For a 1.5 GiB torrent (~6,144 pieces), each bitfield is ~768 bytes. At 6,000 block events/sec with ~4 picker iterations each = **18 MB/sec of allocation churn**.

rqbit's approach: Uses `Arc<Bitfield>` shared references, no cloning.

Fix: Clone once before the loop, or use a shared reference.

Bottleneck Summary — Ranked by Priority

Rank	Bottleneck	Impact	Metric	Code Location
1	End-game pipeline depth=4	+40–60s per download	Throughput	`torrent.rs:5364`
2	Reactive re-requesting cascade	18K picker cycles/sec	Throughput	`torrent.rs:3601`
3	Unbounded store buffer	~50 MiB overhead	Memory	`disk.rs:47`
4	pending_requests O(n) scan	768K comparisons/sec	Throughput	`torrent.rs:3352+`
5	peer_count() O(n log n)	Thousands of sorts/sec	Throughput	`piece_selector.rs:73`
6	Bitfield clones in picker	18 MB/sec alloc churn	Both	`torrent.rs:5375`

Architectural Comparison

Aspect	torrent	rqbit
Pipeline model	128 normal → 4 end-game	Flat 128 always
Block tracking	`Vec` with linear scan	Hash-indexed O(1)
Disk writes	Store buffer → batch verify	Direct `pwritev()`
Bitfield sharing	Clone per picker iteration	`Arc<Bitfield>` references
End-game strategy	Reactive cancel + re-request	Simple dedup, no cascade
Peer state size	~3 KB/peer (Vec + HashSet)	~1 KB/peer (hash maps)

Conclusion

torrent v0.66.0 downloads at **15.0 MB/s** — 3.8x slower than rqbit (56.7 MB/s) and 1.9x slower than qbittorrent (29.1 MB/s). Its only competitive metric is memory vs qbittorrent (170 MiB vs 1.5 GiB).

The **end-game pipeline starvation** (Priority 1) alone accounts for an estimated 40–60 seconds per download. Fixing it — by raising `END_GAME_DEPTH` from 4 to 128 — is the single highest-impact change. Combined with the reactive re-requesting fix (Priority 2) and store buffer eviction (Priority 3), torrent should realistically achieve **40–60 MB/s** and **<80 MiB RSS**, closing the gap with rqbit significantly.